ASCE 41-23 Tier 1 Checklists

FIRM:	
PROJECT NAME:	
SEISMICITY LEVEL:	
PROJECT NUMBER:	
COMPLETED BY:	
DATE COMPLETED:	
REVIEWED BY:	
REVIEW DATE:	

Legend: C = Compliant, NC = Noncompliant, N/A = Not Applicable, U = Unknown

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Project Name _____
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17.13IO Structural Checklist for Building Types C3: Concrete Frames with Infill Masonry Shear Walls and C3a: Concrete Frames with Infill Masonry Shear Walls and Flexible Diaphragms

Table 17-27. Immediate Occupancy Structural Checklists for Building Types C3 and C3a.

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Very Low Seismicity							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	REDUNDANCY: The number of lines of shear walls in each principal direction is greater than or equal to 2.	5.5.1.1	A.3.2.1.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	SHEAR STRESS CHECK: The shear stress in the reinforced masonry shear walls, calculated using the Quick Check procedure of Section 4.4.3.3 is less than 70 lb/in.^{2} (0.48 MPa).	5.5.3.1.1	A.3.2.4.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	SHEAR STRESS CHECK: The shear stress in the unreinforced masonry shear walls, calculated using the Quick Check procedure of Section 4.4.3.3 is less than 30 lb/in.^{2} (0.21 MPa) for clay units and 70 lb/in.^{2} (0.48 MPa) for concrete units. Bays with openings greater than 25% of the wall area shall not be included in A_w of Equation (4-8).	5.5.3.1.1	A.3.2.5.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	INFILL WALL CONNECTIONS: Masonry is in full contact with frame.	5.5.3.5.1 5.5.3.5.3	A.3.2.6.1	
Connections							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	TRANSFER TO SHEAR WALLS: Diaphragms are connected for transfer of loads to the shear walls, and the connections are able to develop the lesser of the shear strength of the walls or diaphragms.	5.7.2	A.5.2.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	CONCRETE COLUMNS: All concrete columns are doweled into the foundation with a minimum of four bars, and the dowels are able to develop the tensile capacity of reinforcement in columns of the seismic-force-resisting system.	5.7.3.1	A.5.3.2	

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Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Low Seismicity (Complete the Following Items in Addition to the Items for Very Low Seismicity)							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	DEFLECTION COMPATIBILITY: Secondary components have the shear capacity to develop the flexural strength of the components and are Compliant with the following items in Table 17-23: COLUMN-BAR SPLICES, BEAM-BAR SPLICES, COLUMN-TIE SPACING, STIRRUP SPACING, and STIRRUP AND TIE HOOKS.	5.5.2.5.2	A.3.1.6.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	FLAT SLABS: Flat slabs or plates not part of the seismic-force-resisting system have continuous bottom steel through the column joints.	5.5.2.5.3	A.3.1.6.3	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	REINFORCING AT WALL OPENINGS: All wall openings that interrupt rebar have trim reinforcing on all sides.	5.5.3.1.5	A.3.2.4.3	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	PROPORTIONS: The height-to-thickness ratio of the unreinforced infill walls at each story is less than 13.	5.5.3.1.2	A.3.2.6.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	CAVITY WALLS: The infill walls are not of cavity construction.	5.5.3.5.2	A.3.2.6.3	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	INFILL WALLS: The infill walls are continuous to the soffits of the frame beams and to the columns to either side.	5.5.3.5.3	A.3.2.6.4	
Diaphragms (Stiff or Flexible)							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	DIAPHRAGM CONTINUITY: Floor and roof diaphragms do not have expansion joints or vertical offsets, such as split levels, sawtooth, or clerestory configurations.	5.6.1.1	A.4.1.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	OPENINGS AT SHEAR WALLS: Diaphragm openings immediately adjacent to the shear walls are less than 15% of the wall length.	5.6.1.3	A.4.1.4	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	OPENINGS AT EXTERIOR MASONRY SHEAR WALLS: Diaphragm openings immediately adjacent to exterior masonry shear walls are not greater than 4 ft (1.2 m) long.	5.6.1.3	A.4.1.6	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	PLAN IRREGULARITIES: There is tensile capacity to develop the strength of the diaphragm at reentrant corners or other locations of plan irregularities.	5.6.1.4	A.4.1.7	

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C	NC	N/A	U			
☐	☐	☐	☐	DIAPHRAGM REINFORCEMENT AT OPENINGS: There is reinforcing around all diaphragm openings larger than 50% of the building width in either major plan dimension.	5.6.1.5	A.4.1.8
Flexible Diaphragms						
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	CROSSTIES: There are continuous crossties between diaphragm chords.	5.6.1.2	A.4.1.2
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	STRAIGHT SHEATHING: All straight-sheathed diaphragms have horizontal spans less than 12 ft (3.6 m) and aspect ratios less than 1-to-1 in the direction being considered.	5.6.2	A.4.2.1
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	DIAGONALLY SHEATHED AND UNBLOCKED DIAPHRAGMS: All diagonally sheathed or unblocked wood structural panel diaphragms have horizontal spans less than 30 ft (9.1 m) and aspect ratios less than or equal to 3-to-1.	5.6.2	A.4.2.2
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	BLOCKED DIAPHRAGMS: All blocked wood structural panel diaphragms have horizontal spans less than 90 ft (27.4 m) and have aspect ratios less than or equal to 4-to-1.	5.6.2	A.4.2.3
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	CANTILEVERED WOOD DIAPHRAGMS: All cantilevered wood diaphragms that provide lateral support for concrete or masonry walls consist of wood structural panels and have a maximum cantilever length of 15 ft (4.6 m) if unblocked or 25 ft (7.6 m) if blocked, and a maximum ratio of cantilever length to diaphragm width of 1:2.5 if unblocked and 1:1.5 if blocked. In addition, the cantilevered diaphragm has a back-span length equal to or greater than the cantilevered portion.	5.6.2	A.4.2.4
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	NON-CONCRETE-FILLED DIAPHRAGMS: Bare steel deck diaphragms or steel deck diaphragms with fill other than reinforced concrete consist of horizontal spans of less than 40 ft (12.2 m) and have aspect ratios less than 4-to-1.	5.6.3	A.4.3.1

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C	NC	N/A	U			
☐	☐	☐	☐	OTHER DIAPHRAGMS: Diaphragms do not consist of a system other than wood, steel deck, concrete, or horizontal bracing.	5.6.5	A.4.7.1
Connections						
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	UPLIFT AT PILE CAPS: Pile caps have top reinforcement, and piles are anchored to the pile caps; the pile cap reinforcement and pile anchorage are able to develop the tensile capacity of the piles.	5.7.3.5	A.5.3.8
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	STIFFNESS OF WALL ANCHORS: Anchors of concrete or masonry walls to wood structural elements are installed taut and are stiff enough to limit the relative movement between the wall and the diaphragm to no greater than 1/8 in. (3.1 mm) before engagement of the anchors.	5.7.1.2	A.5.1.4

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Moderate Seismicity (Complete the Following Items in Addition to the Items for Very Low and Low Seismicity)							
Foundation System							
C	**NC**	**N/A**	**U**				
☐	☐	☐	☐	OVERTURNING: The ratio of the least horizontal dimension of the seismic-force-resisting system at the foundation level to the building height (base/height) is greater than $0.6S_a$.	5.4.3.3	A.6.2.1	

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
High Seismicity (Complete the Following Items in Addition to the Items for Low and Moderate Seismicity)							
Seismic-Force-Resisting System							
C	**NC**	**N/A**	**U**				
☐	☐	☐	☐	PROPORTIONS: The height-to-thickness ratio of the unreinforced infill walls at each story is less than 8.	5.5.3.1.2	A.3.2.6.2	

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